# Mastering The Linux Command Line for Ethical Hacking

Akintunde Ridwan Oladimeji
**Student ID** : 2025/INT/11578

**Course Name** : Linux Operating System Fundamentals
**Instructor Name** : Mr Aminu Idris (AMCPN)

**Date of Submission** : 13/12/2025
**Version :** 1.0

**Table of Content**


Executive Summary

1.0 Introduction

2.0 Aim & Objectives

2.1 Tools and Resources Used

3.0 Methodology

3.1 Creating a complex directory

3.2 Setting permission on different files for access control

3.2.1 All ".sh" files made executable by the owner

3.2.2 All ".txt" files made readable by everyone but only writable by owner

3.2.3 Creating an inaccessible file for User, Group and Organisation

3.3 Locating files with keywords and extensions for various operations

4.0 Analysis and Finding

5.0 Challenges and Solution

6.0 Conclusion

7.0 Recommendation

9.0 References

**Executive Summary**

This Lab focused on exploring and mastering the linux command line for ethical hacking. The key activities include creating directory and file structure for penetration test engagement, setting up permission schemes on different files for access control, locating files with keywords and extensions for various operations and directing results from an operation to another as an input using different linux commands. This task justified the uniqueness and vast opportunities associated with mastering linux command for carrying.out different operations in ethical hacking.

## 1.0 Introduction

Linux is a free, open-source operating system known for its flexibility, stability, and strong security. It is widely used in personal computing, server environments, and enterprise systems because of its performance and customization capabilities.

Kali Linux is a Debian-derived Linux distribution designed for digital forensics and penetration testing. It comes pre-installed with a vast collection of security tools, making it the top choice for security professionals.

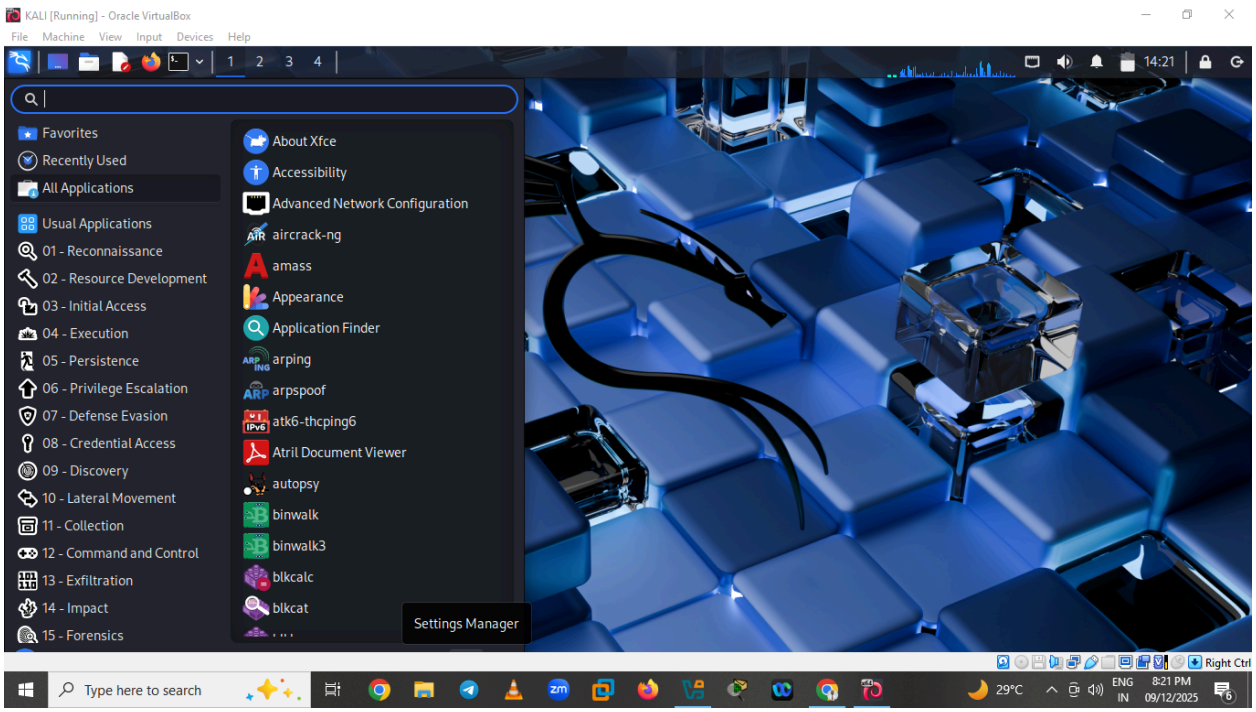


**Fig 1**: kali Linux interface showing preinstalled app and packages for ethical hacking and digital forensics operations.

The command line interface (CLI) is a crucial part of working with Kali Linux, as most of its powerful tools and utilities operate without a Graphical User Interface (GUI). When performing ethical hacking, penetration testing, or system administration, you'll spend most of your time interacting directly with the terminal.

In Kali Linux, commands are executed through the terminal emulator, where users enter specific instructions.

After execution, the terminal returns the corresponding output or system response, providing precise control over tools, automation, and overall system operations.

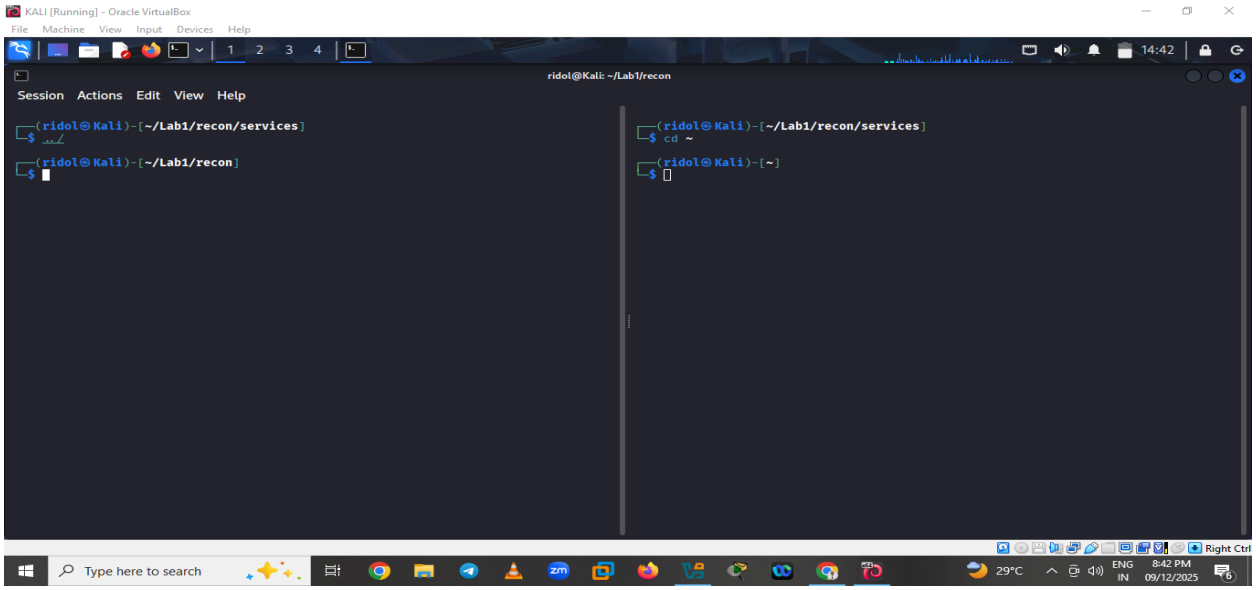


**Fig 2:** kali Linux command Line interface showing a splitted screen view for multiple operations.

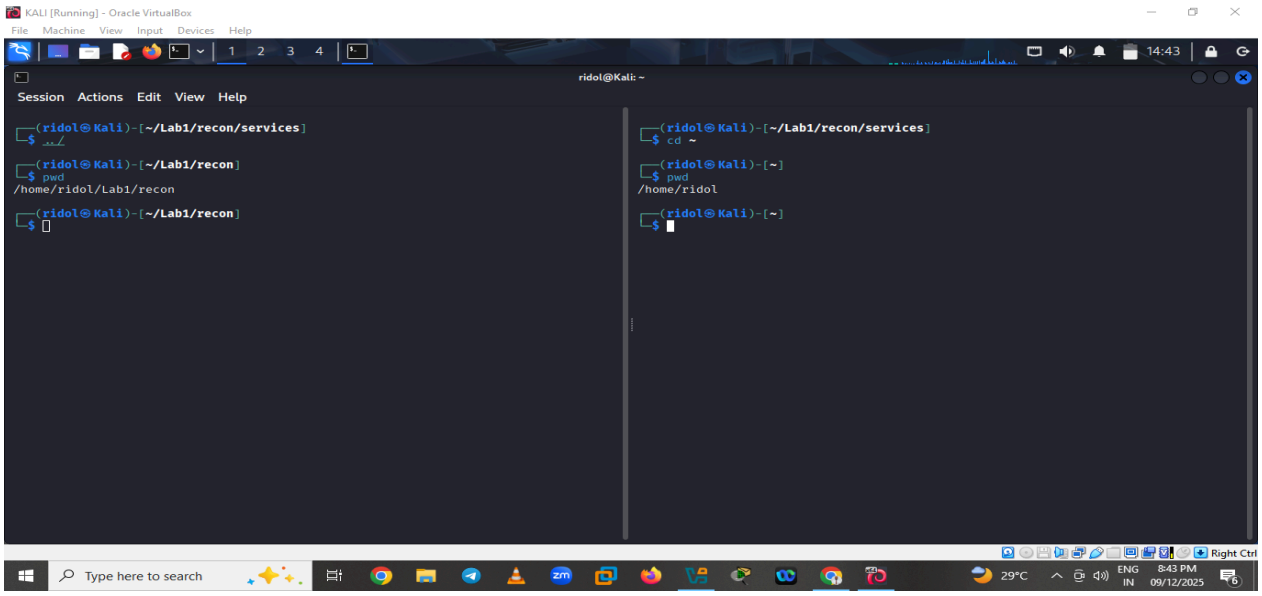


**Fig 3:** Linux Terminal showing basic command operations such as ***whoami ,pwd, ls -l, cd ~,***

The interesting part of exploring Kali Linux tends to be incredible when these basic commands are blended together to form or carried out sophisticated task at a go

Such as - ***mkdir -p LAb1/{recon/{hosts, services), exploits, logs, evidence/{screenshots, packets}}***

This created a directory, subdirectory and sub folder.

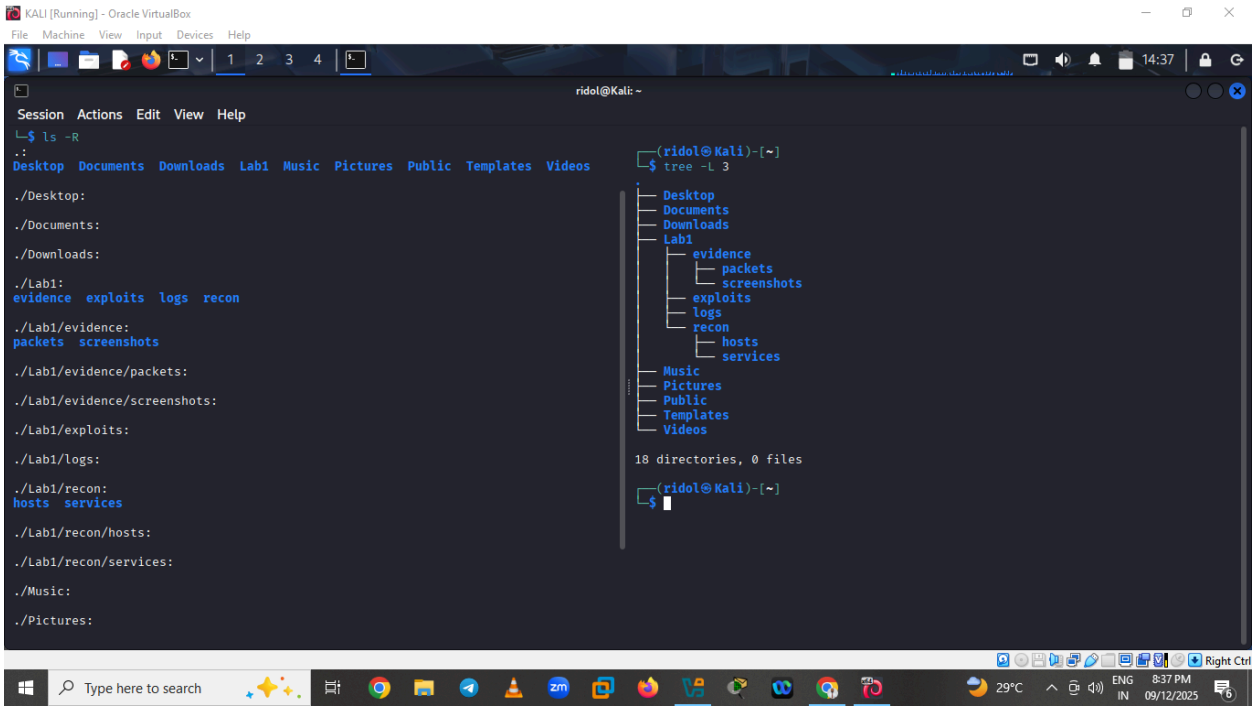


**Fig 4:** Linux Terminal showing Directory and subdirectory created with one - ***mkdir -p LAb1/{recon/{hosts, services), exploits, logs, evidence/{screenshots, packets}}*** operation.

## 2.0 Aim & Objectives

The aim of this lab is to explore the Kali linux interface and it commands while carrying out various operations with these commands

The primary objective of this lab is

- to create directory and file structure for penetration test engagement
- to set up permission scheme on different files for access control
- to locate files with keywords and extensions for various operations .
- to direct result from an operation to another file as an input

## 2.1 Tools and Resources Used

Kali Linux : A penetration testing platform used for security assessment

## 3.0 Methodology

## 3.1 Creating a complex directory

A complex directory was created on the kali linux machine containing the main directory, subdirectories, subfolders and files in a single chain of command.

Steps:

- Opened the kali Linux command line interface
- Ran the command ***mkdir -p*** *Engagement_2024/{reconnaissance,exploitation,reporting,evidence/{screenshot,logs,packets}}*
- The command ***mkdir*** along with the argument passed created a main directory and subdirectories all together at once
- Ran the command ***touch*** *{reconnaissance,exploitation,reporting,evidence} /readme.txt*
- The command ***touch*** created a readme.txt file in all the subdirectories at once
- Verified the created directory,subdirectories and subfolders with ***ls -R*** or ***tree*** command

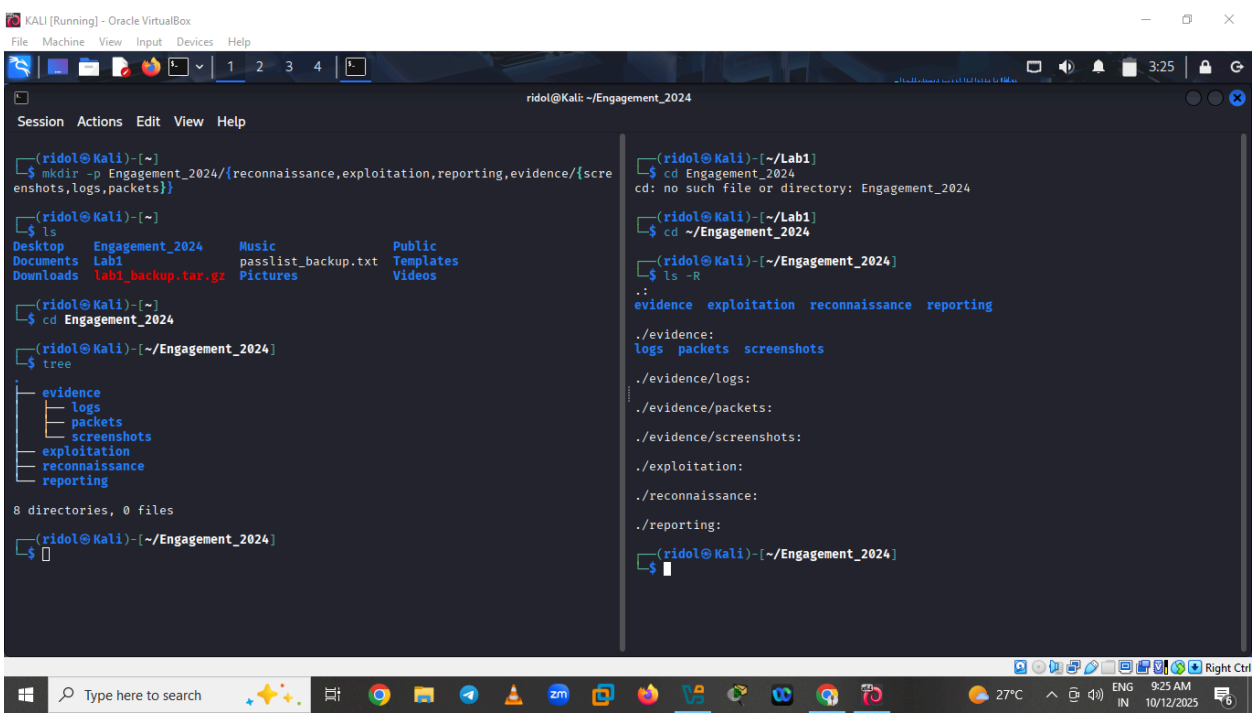


**Fig 5:** Linux Terminal showing evidence of directory and subdirectories created in a single step.

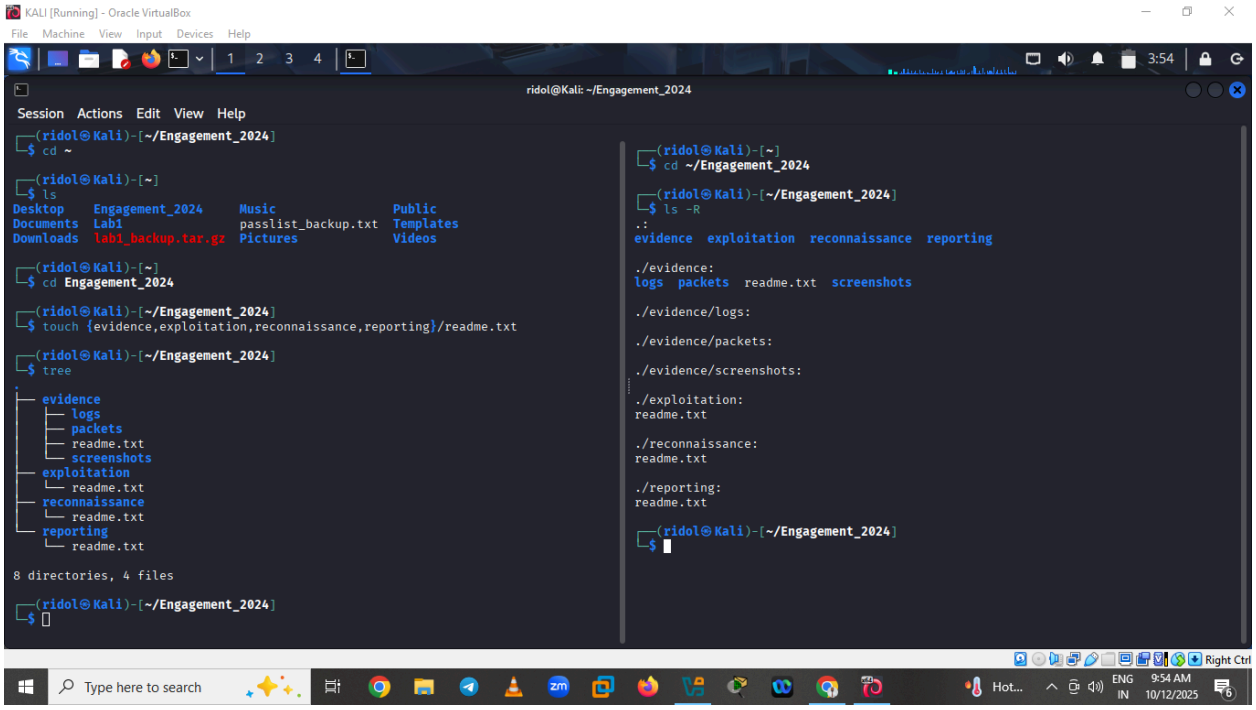


**Fig 6:** Linux Terminal showing evidence of ***readme.txt*** file created in all directories in a single step.

## 3.2 Setting permission on different files for access control

### 3.2.1 All ".sh" files made executable by the owner

- Opened Kali Linux command line interface
- Ran the command ***find -name "*.sh"*** to display all files with ***"*.sh"*** extension
- Ran the command ***find -name "*.sh" -exec chmod 766 {} \;***
- The "7" gave the User full permission to **Read+Write+Execute**
- The "6" gave the Group permission to **Read+Write only**
- The "6" gave the Organization permission to **Read+Write only**

- Verified the permission effect with the command ***find -name /home/ridol -type f -name "*.sh" -exec ls -l {} \;***
- The command carried out a search process for files with the name specified and passed the result to the next command which displayed the file permission for the obtained result.

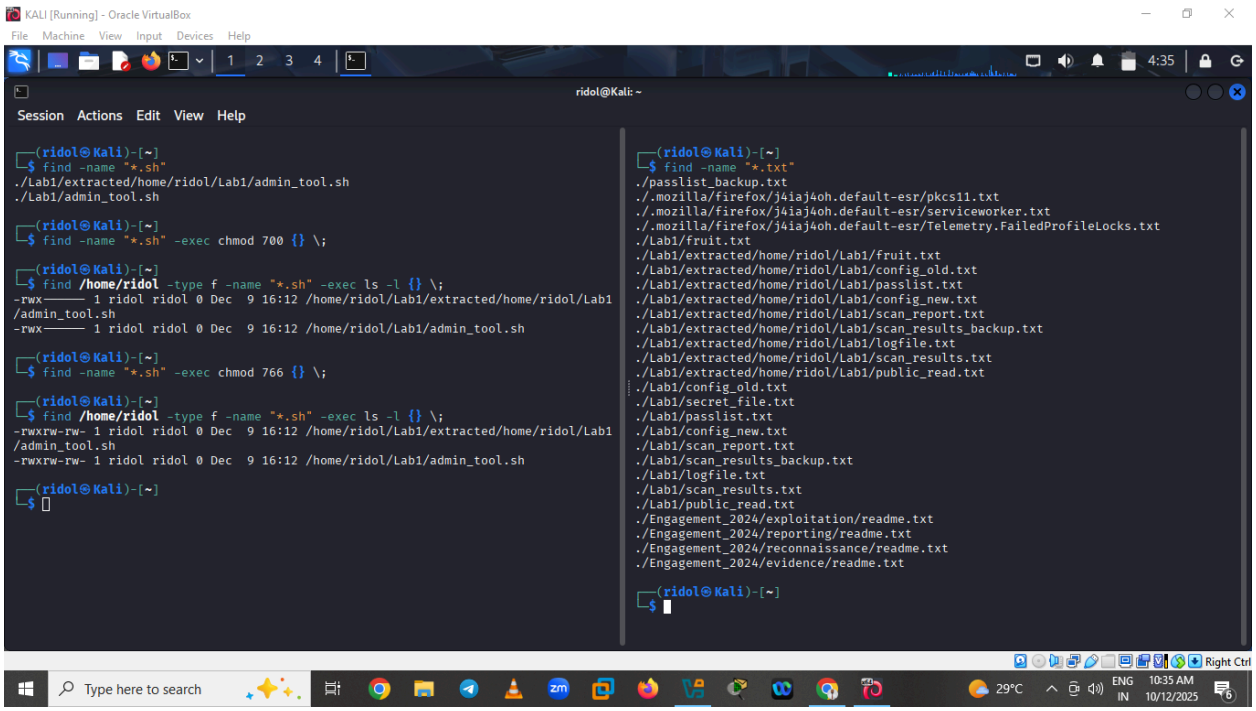


**Fig 7:** Linux Terminal showing evidence of all "***.sh***" files made executable by the owner

### 3.2.2 All ".txt" files made readable by everyone but only writable by owner

- Opened Kali Linux command line interface
- Ran the command ***find -name "*.txt"*** to display all files with ***"*.txt"*** extension
- Ran the command ***find -name "*.txt" -exec chmod 644 {} \;***
- The "6" gave the User permission to **Read+Write only**
- The "4" gave the Group permission to **Read only**
- The "4" gave the Organization permission to **Read only**
- Verified the permission effect with the command ***find -name /home/ridol -type f -name "*.txt" -exec ls -l {} \;***

- The command carried out a search process for files with the name specified and passed the result to the next command which displayed the file permission for the obtained result

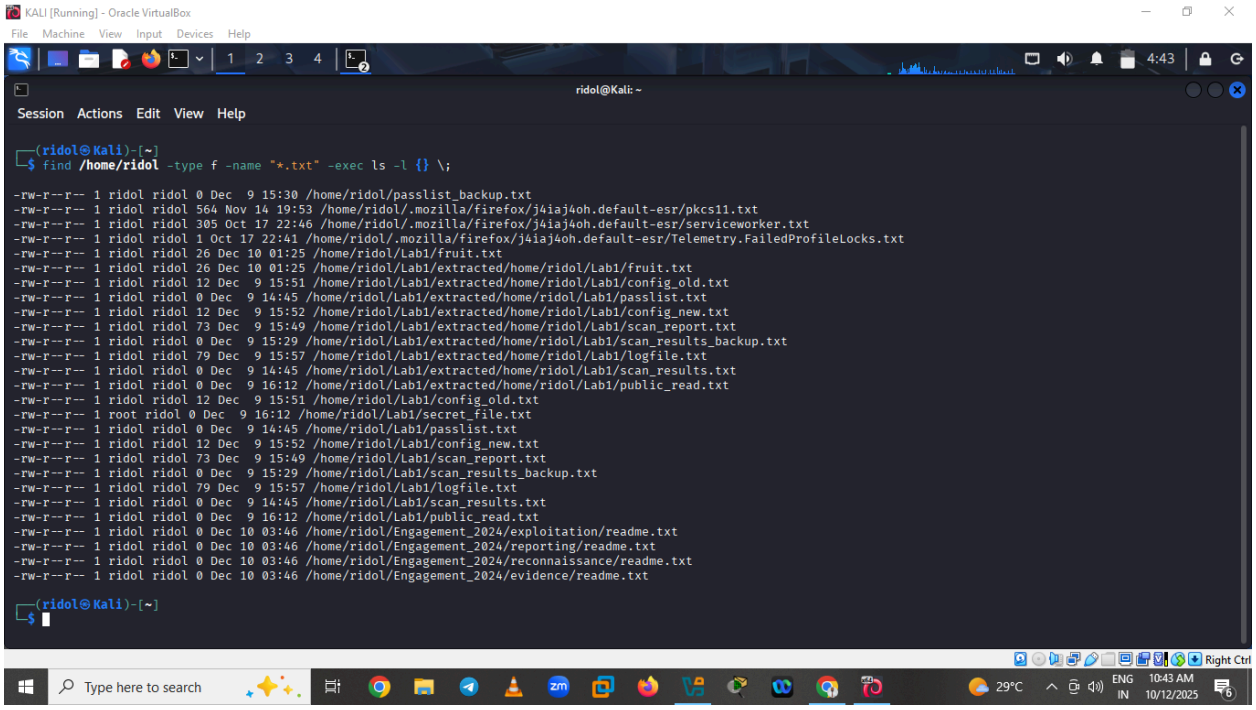


**Fig 8:** Linux Terminal showing evidence of all "***.txt***" files made readable by everyone but only writable by owner

### 3.2.3 Creating an inaccessible file for User, Group and Organisation

- Opened Kali Linux command line interface
- Ran the command ***touch Unauthorized_file*** to create a file named Unauthorized_file
- Ran the command ***chmod 000 Unauthorized_file***
- The "0" gave the User,Group and Organisation zero access to neither read,write nor execute the file
- The file can only be accessed with a SuperUser privilege (**Sudo**)
- Verified the permission effect with the command ***find -name /home/ridol -type f -name "Unauthorized_file" -exec ls -l {} \;***

- The command carried out a search process for files with the name specified and passed the result to the next command which displayed the file permission for the obtained result.

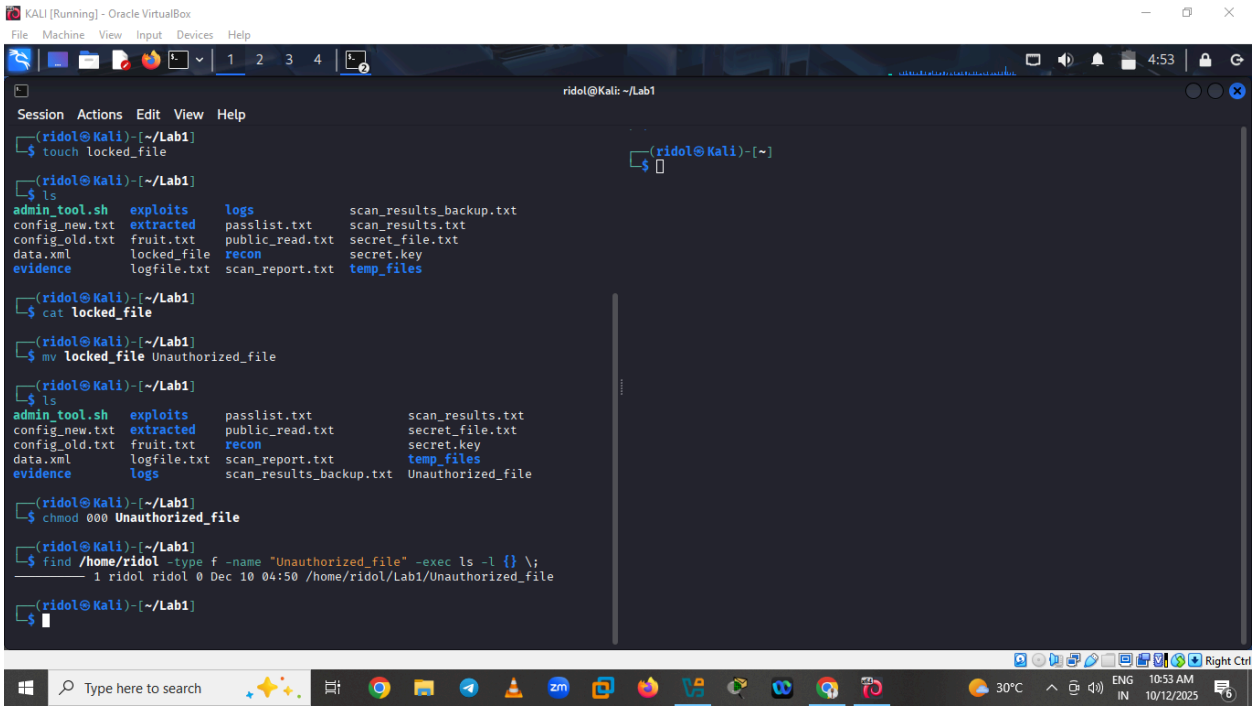


**Fig 9:** Linux Terminal showing evidence of a created file "***Unauthorized_file***" file made inaccessible by everyone.

## 3.3 Locating files with keywords and extensions for various operations

- Opened Kali Linux command line interface
- Ran the command ***find -name "*.log"*** to display all files with ***"*.log"*** extension
- Ran the command ***find -name "*.log" -exec grep -R "ERROR" {} \;*** to display all files with ***"*.log"*** extension that contain keyword **"ERROR"** and their recursive path
- Ran the command ***grep -R "ERROR"*** which exposed the path and actual files that contains the keywords **"ERROR"** as **"logfile.txt"**
- Ran the command ***find -name "*logfile.txt" -exec grep -R "ERROR" {} \;*** to display all files with ***"*logfile.txt"*** extension that contain keyword **"ERROR"** and their recursive path

- The command carried out a search process for files using their extension or keywords and located their path with the help of the recursive argument.
- Ran the command ***Sudo grep -R "ERROR" | wc -l*** to carry out word count (lines with ERROR in all the logfile.txt search
- Ran the command ***grep -R "ERROR" ~ > error_report.txt*** to export the search result into a file (error_report.txt)
- Verified the result by issuing a ***cat error_report.txt*** command

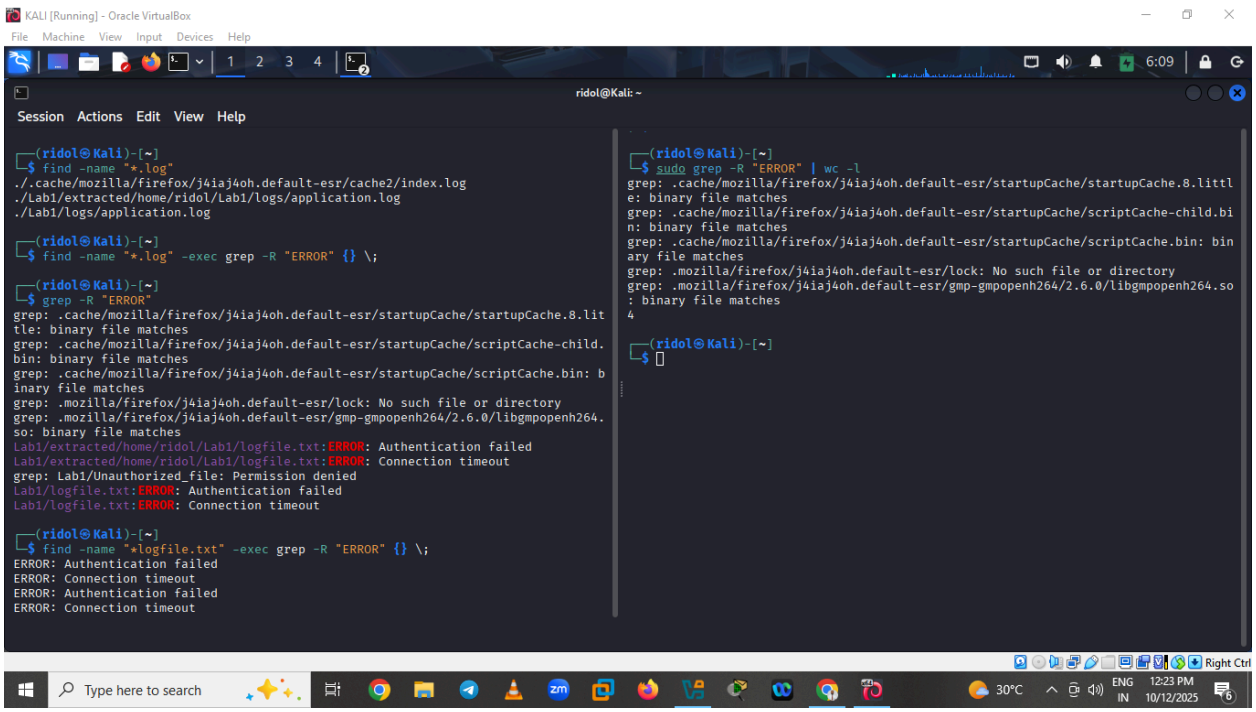


**Fig 10:** Linux Terminal showing evidence of all "***.log***" and "***ERROR***" keywords search result.

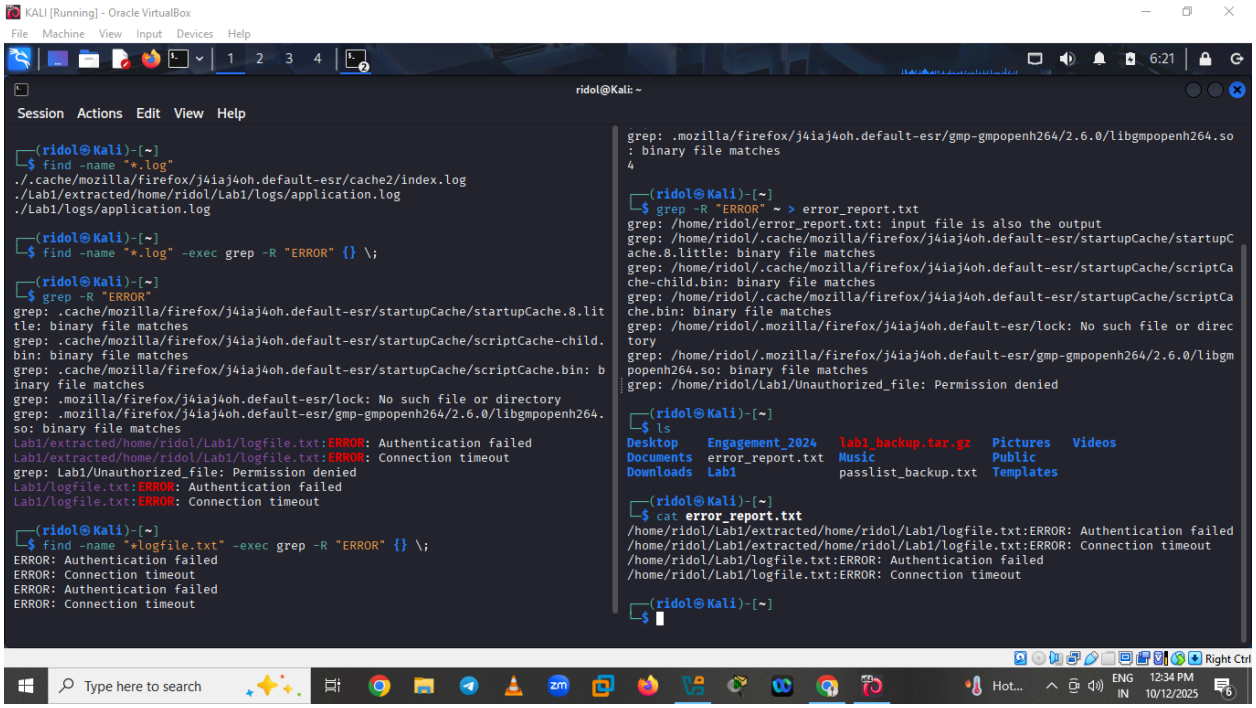


**Fig 11:** Linux Terminal showing evidence of all lines with "***ERROR***" keywords search results and exporting the result to "***error_report.txt***" file.

## 4.0 Analysis and Finding

The command line interface exploration with the creation of complex directory structure with subdirectories and files in a single chained command justified the power of kali linux in executing operation effectively and efficiently thereby saving time and optimising operational bottleneck.

The file permission with **766** strictly gave the user complete access and permission to execute the file while limiting other groups and organizations to read and write access only.
Similarly, the file permission with **644** limits the user to read and write permission alone while limiting other groups and organizations to read access only.
Contrary to the two file permission types, a file with permission of **000** will be inaccessible for both user, group and organisation and will require a superuser (sudo) privilege to access.

The **grep** and **find** command play a very vital role in locating files when combined with other commands and arguments such as **type -R** which could locate files either by keywords, types or extension.
The **.log** files were found with the **find** command but contains no **ERROR** words.The **grep** command exposed the files **logfile.txt** which happens to contain the keywords **ERROR** in question.
The output and result from one operation can be directly exported into another file or process with the **>** flag just as it was done in ***grep -R "ERROR" ~> error_report.txt*** scenario.

## 5.0 Challenges and Solution

Challenges encountered basically include syntax error while trying to chain two or more commands together while trying out new complex commands outside the given ones.
Most of the tried commands eventually worked after careful review and correction of the syntax error flagged.

## 6.0 Conclusion

This lab introduced, refreshed and scaled up the vital fundamental linux command line skills needed for penetration testing to stay at the forefront of the industry.

## 7.0 Recommendation

To attain mastery in linux command line for ethical hacking and penetration testing students and aspiring ethical hacker should try and explore the relationship between several command line and how they can perform complex operational tasks when combined together beyond their usual individual capacity

## 9.0 References

International Cybersecurity and Digital Forensics Academy (2025). *Mastering Linux Command Line for Ethical Hacking*